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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2025-2026

IE2106 – ENGINEERING MATHEMATICS I

April / May 2026 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 12 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.
6. A list of useful formulae is given in Appendix A on pages 4-5, followed by Statistical Tables on pages 6-10, and the Laplace Transform Table on pages 11-12.

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1. The R&D division of a Japanese car manufacturer carries out tests on its newest hybrid model *Icebird*. From a random sample of 30 Icebird cars, it is recorded that the fuel consumption rate has a mean of 33 km/litre, and a standard deviation of 0.6 km/litre.

   (a) Construct a 98% confidence interval for the standard deviation of the fuel consumption rate of Icebird. State any assumptions/requirements needed. Does the 98% confidence interval contain the standard deviation? Explain.

   (7 Marks)

   (b) From a market survey conducted in December 2025, the preference of hybrid car enthusiasts is to have the average fuel consumption rate above $f_0 = 32.5$ km/litre.

      (i) From the sample of 30 Icebird cars, can the car manufacturer conclude, at the 5% level of significance, that Icebird will be a hit with the hybrid car enthusiasts? State any assumptions/requirements needed.

Note: Question No. 1 continues on page 2

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(ii) The number f_0 may vary for subsequent market surveys. The car manufacturer is concerned that Icebird may be out of favour with the hybrid car enthusiasts in the future. What is the range of f_0 that will ensure the support of the hybrid car enthusiasts for Icebird at 5% level of significance?

(10 Marks)

(c) After many tests and measurements, suppose the standard deviation of the fuel consumption rate of Icebird is now known as 0.63 km/litre. The R&D division aims to improve the fuel consumption rate of Icebird in the near future. To ascertain the degree of improvement through testing, what range of *sample mean of fuel consumption rate* should they target so as to ensure Icebird is a hit with the hybrid car enthusiasts at 2.5% level of significance? Assume that the sample size is still 30 and f_0 is elevated to 33.1 km/litre.

(8 Marks)

2. (a) Let X be the number of seeds in an apple. X has the probability mass function

$$P(X=4)=0.3,\quad P(X=5)=0.25,\quad P(X=6)=c,\quad P(X=7)=d,\quad P(X=8)=0.05,$$

where c and d are constants. It is known that $E(X^2)=29.95$. Find c and d. Hence, compute $E(2-3X)$ and $Var(2-3X)$.

(11 Marks)

(b) The number of traffic accidents occurring at a particular road junction in a month can be modelled by a Poisson distribution. On average, 0.4 accidents occur per month.

(i) Find the probability that there are at least 2 accidents occurring at the road junction within a year. Give your answer correct to 3 decimal places.

(ii) The Land Transport Department (LTD) is monitoring this particular road junction on a yearly basis. Till April this year, 2 accidents at the road junction have been recorded. What is the probability that within this year there are at most 4 accidents happening at this road junction? Give your answer correct to 3 decimal places.

(iii) Let λ be the mean rate of accidents happening at the road junction in a year. LTD would like to improve the situation and set a target of achieving probability of at least 0.5 that zero accident happens at this road junction in a year. What λ should LTD be aiming for?

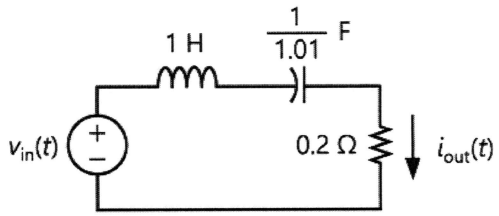
(14 Marks)

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3. (a) The schematic shown below represents a band-pass filter comprising an inductor of 1 H, a capacitor of $\frac{1}{1.01}$ F, and a resistor of 0.2 Ω.

(i) Derive the transfer function in the Laplace domain ($H(s) = \frac{I_{\text{out}}(s)}{V_{\text{in}}(s)}$).

(ii) Determine $i_{\text{out}}(t)$ when the input $v_{\text{in}}(t)$ is a delta function ($v_{\text{in}}(t) = \delta(t)$).



(15 Marks)

(b) Using Newton's divided difference interpolation method, derive the second-degree polynomial that interpolates $f(x) = \frac{1}{x}$ at points $x_0 = 1$, $x_1 = 2$, and $x_2 = 3$.

(10 Marks)

4 (a) Use the Laplace transform technique to solve the differential equation

$$2y'' + y' = 1, \qquad t \geq 0$$

subject to the initial conditions $y(0) = 1$ and $y'(0) = 0$.

(15 Marks)

(b) Find positive constants a and b such that $u(x,t) = e^{-6t}(e^{ax} + e^{-bx})$ satisfies the partial differential equation

$$u_t + u_x + u_{xx} = 0.$$

(10 Marks)

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Appendix A

Total Probability of A

$$P(A) = \sum_{j=1}^{k} P(A \mid B_j) P(B_j)$$

Bayes Theorem

$$P(B_r \mid A) = \frac{P(A \mid B_r) P(B_r)}{\sum_{j=1}^{k} P(A \mid B_j) P(B_j)}$$

Poisson Distribution

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

Binomial Distribution

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Fixed-Point Iteration Method

$$x_{n+1} = g(x_n), \quad n = 0, 1, 2, \ldots$$

Euler Method

$$y_{n+1} = y_n + hf(x_n, y_n)$$

Improved Euler Method (Heun's Method)

$$a = hf(x_n, y_n)$$
$$b = hf(x_{n+1}, y_n + a)$$
$$y_{n+1} = y_n + \frac{1}{2}(a + b)$$

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Appendix A (Continued)

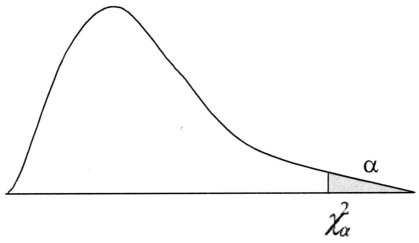
Runge-Kutta Method

$$a = h\,f(x_n, y_n)$$
$$b = h\,f(x_n + \tfrac{1}{2}h,\ y_n + \tfrac{1}{2}a)$$
$$c = h\,f(x_n + \tfrac{1}{2}h,\ y_n + \tfrac{1}{2}b)$$
$$d = h\,f(x_n + h,\ y_n + c)$$
$$y_{n+1} = y_n + \tfrac{1}{6}(a + 2b + 2c + d)$$

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Statistical Table - 1

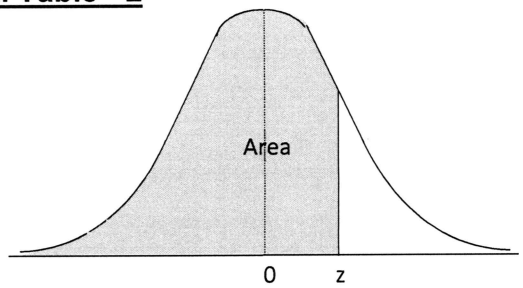
Percentage Points of χ^2 Distribution



d.f. \ α	.99	.975	.95	.90	.50	.10	.05	.025	.01
1	.0002	.001	.004	.02	.45	2.71	3.84	5.02	6.63
2	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21
3	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34
4	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28
5	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09
6	.87	1.24	1.64	2.20	5.35	10.64	12.59	14.45	16.81
7	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48
8	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09
9	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67
10	2.56	3.24	3.94	4.87	9.34	15.99	18.31	20.48	23.21
11	3.05	3.81	4.57	5.58	10.34	17.28	19.68	21.92	24.72
12	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22
13	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69
14	4.66	5.62	6.57	7.79	13.34	21.06	23.68	26.12	29.14
15	5.23	6.26	7.26	8.55	14.34	22.31	25.00	27.49	30.58
16	5.81	6.90	7.96	9.31	15.34	23.54	26.30	28.85	32.00
17	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41
18	7.01	8.23	9.39	10.86	17.34	25.99	28.87	31.53	34.81
19	7.63	8.90	10.12	11.65	18.34	27.20	30.14	32.85	36.19
20	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57
21	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93
22	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29
23	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64
24	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98
25	11.52	13.11	14.61	16.47	24.34	34.38	37.65	40.65	44.31
26	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64
27	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96
28	13.56	15.30	16.93	18.94	27.34	37.92	41.34	44.46	48.28
29	14.26	16.04	17.71	19.77	28.34	39.09	42.56	45.72	49.59
30	14.95	16.78	18.49	20.60	29.34	40.26	43.77	46.98	50.89
40	22.16	24.42	26.51	29.05	39.34	51.81	55.76	59.34	63.69
50	29.71	32.35	34.76	37.69	49.33	63.17	67.50	71.42	76.15
60	37.48	40.47	43.19	46.46	59.33	74.40	79.08	83.30	88.38
70	45.44	48.75	51.74	55.33	69.33	85.53	90.53	95.02	100.43
80	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33
90	61.75	65.64	69.13	73.29	89.33	107.57	113.15	118.14	124.12
100	70.06	74.22	77.93	82.36	99.33	118.50	124.34	129.56	135.81

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Statistical Table - 2



Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0352	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0722	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Note: Statistical Table 2 continues on page 8.

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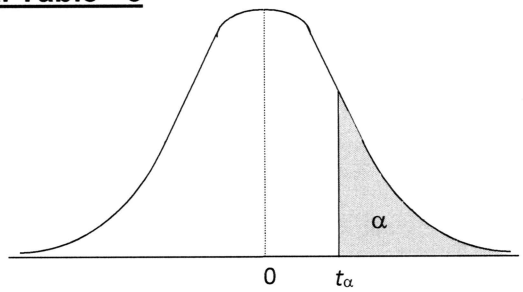
Statistical Table – 2 (Continued)

Areas Under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9278	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

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Statistical Table - 3



Critical Values of the *t*-Distribution

	α						
ν	0.40	0.30	0.20	0.15	0.10	0.05	0.025
1	0.325	0.727	1.376	1.963	3.078	6.314	12.706
2	0.289	0.617	1.061	1.386	1.886	2.920	4.303
3	0.277	0.584	0.978	1.250	1.638	2.353	3.182
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228
11	0.260	0.540	0.876	1.088	1.363	1.796	2.201
12	0.259	0.539	0.873	1.083	1.356	1.782	2.179
13	0.259	0.537	0.870	1.079	1.350	1.771	2.160
14	0.258	0.537	0.868	1.076	1.345	1.761	2.145
15	0.258	0.536	0.866	1.074	1.341	1.753	2.131
16	0.258	0.535	0.865	1.071	1.337	1.746	2.120
17	0.257	0.534	0.863	1.069	1.333	1.740	2.110
18	0.257	0.534	0.862	1.067	1.330	1.734	2.101
19	0.257	0.533	0.861	1.066	1.328	1.729	2.093
20	0.257	0.533	0.860	1.064	1.325	1.725	2.086
21	0.257	0.532	0.859	1.063	1.323	1.721	2.080
22	0.256	0.532	0.858	1.061	1.321	1.717	2.074
23	0.256	0.532	0.858	1.060	1.319	1.714	2.069
24	0.256	0.531	0.857	1.059	1.318	1.711	2.064
25	0.256	0.531	0.856	1.058	1.316	1.708	2.060
26	0.256	0.531	0.856	1.058	1.315	1.706	2.056
27	0.256	0.531	0.855	1.057	1.314	1.703	2.052
28	0.256	0.530	0.855	1.056	1.313	1.701	2.048
29	0.256	0.530	0.854	1.055	1.311	1.699	2.045
30	0.256	0.530	0.854	1.055	1.310	1.697	2.042
40	0.255	0.529	0.851	1.050	1.303	1.684	2.021
60	0.254	0.527	0.848	1.045	1.296	1.671	2.000
120	0.254	0.526	0.845	1.041	1.289	1.658	1.980
∞	0.253	0.524	0.842	1.036	1.282	1.645	1.960

Note: Statistical Table 3 continues on page 10.

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Statistical Table - 3 (Continued)

Critical Values of the *t*-Distribution

	α						
ν	0.02	0.015	0.01	0.0075	0.005	0.0025	0.0005
1	15.895	21.205	31.821	42.434	63.657	127.322	636.590
2	4.849	5.643	6.965	8.073	9.925	14.089	31.598
3	3.482	3.896	4.541	5.047	5.841	7.453	12.924
4	2.999	3.298	3.747	4.088	4.604	5.598	8.610
5	2.757	3.003	3.365	3.634	4.032	4.773	6.869
6	2.612	2.829	3.143	3.372	3.707	4.317	5.959
7	2.517	2.715	2.998	3.203	3.499	4.029	5.408
8	2.449	2.634	2.896	3.085	3.355	3.833	5.041
9	2.398	2.574	2.821	2.998	3.250	3.690	4.781
10	2.359	2.527	2.764	2.932	3.169	3.581	4.587
11	2.328	2.491	2.718	2.879	3.106	3.497	4.437
12	2.303	2.461	2.681	2.836	3.055	3.428	4.318
13	2.282	2.436	2.650	2.801	3.012	3.372	4.221
14	2.264	2.415	2.624	2.771	2.977	3.326	4.140
15	2.249	2.397	2.602	2.746	2.947	3.286	4.073
16	2.235	2.382	2.583	2.724	2.921	3.252	4.015
17	2.224	2.368	2.567	2.706	2.898	3.222	3.965
18	2.214	2.356	2.552	2.689	2.878	3.197	3.922
19	2.205	2.346	2.539	2.674	2.861	3.174	3.883
20	2.197	2.336	2.528	2.661	2.845	3.153	3.849
21	2.189	2.328	2.518	2.649	2.831	3.135	3.819
22	2.183	2.320	2.508	2.639	2.819	3.119	3.792
23	2.177	2.313	2.500	2.629	2.807	3.104	3.768
24	2.172	2.307	2.492	2.620	2.797	3.091	3.745
25	2.167	2.301	2.485	2.612	2.787	3.078	3.725
26	2.162	2.296	2.479	2.605	2.779	3.067	3.707
27	2.158	2.291	2.473	2.598	2.771	3.057	3.690
28	2.154	2.286	2.467	2.592	2.763	3.047	3.674
29	2.150	2.282	2.462	2.586	2.756	3.038	3.659
30	2.147	2.278	2.457	2.581	2.750	3.030	3.646
40	2.125	2.250	2.423	2.542	2.704	2.971	3.551
60	2.099	2.223	2.390	2.504	2.660	2.915	3.460
120	2.076	2.196	2.358	2.468	2.617	2.860	3.373
∞	2.054	2.170	2.326	2.432	2.576	2.807	3.291

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Laplace Transform Table

		$f(t)$	$F(s)$
1.	Unit impulse	$\delta(t)$	1
2.	Unit step	$u(t)$	$\frac{1}{s}$
3.	Unit ramp	$r(t) = tu(t)$	$\frac{1}{s^2}$
4.	Unit parabola	$p(t) = \frac{1}{2}t^2u(t)$	$\frac{1}{s^3}$
5.	Exponential	e^{-at}	$\frac{1}{s+a}$
6.	t-Multiplication exponential	te^{-at}	$\frac{1}{(s+a)^2}$
7.	Sine	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
8.	Cosine	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
9.	Damped Sine	$e^{-at}\sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
10.	Damped Cosine	$e^{-at}\cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
11.	Linearity	$c_1f_1(t)+c_2f_2(t)$	$c_1F_1(s)+c_2F_2(s)$
12.	Differentiation	$\frac{d}{dt}f(t)$	$sF(s)-f(0)$
13.	n-Fold differentiation	$\frac{d^n}{dt^n}f(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)$ $-\cdots\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$
14.	Integration	$\int_{0-}^{t} f(\tau)d\tau$	$\frac{F(s)}{s}$
15.	t-Multiplication	$tf(t)$	$\frac{-d}{ds}F(s)$

Note: Laplace Transform Table continues on page 12.

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Laplace Transform Table (Continued)

	$f(t)$	$F(s)$
16. n-Fold t-Multiplication	$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n} F(s)$
17. Time shift	$f(t-t_0)u(t-t_0)$; $t_0 > 0$	$e^{-st_0} F(s)$
18. Frequency shift	$e^{-s_0 t} f(t)$	$F(s+s_0)$
19. Time-frequency scaling	$f(ct)$; $c > 0$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
20. Periodic Function	$f(t) = f_1(t)u(t) + f_1(t-T)u(t-T) + f_1(t-2T)u(t-2T) + \cdots$	$\frac{1}{1-e^{-sT}} F_1(s)$

END OF PAPER

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IE2106 ENGINEERING MATHEMATICS I

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3. Please write your Matriculation Number on the front of the answer book.

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